NASA SPACE APPS CHALLENGE 2026

Astro Sağlık

Space Station Health Mission

A 3D game where kids monitor an astronaut's health, decide what is wrong and act fast to keep the crew safe.

Team: AstroPulse

Member: Azra Kurt (solo team)

Challenge: Create Health Monitoring Software for Astronauts on Space Missions



Project Demo · 7 slides

The problem: space is hard on the human body

NASA's Human Research Program names five hazards of human spaceflight (RIDGE):



Radiation

Higher cancer & tissue risk



Isolation

Stress, mood, team conflict



Distance

No quick trip to a doctor



Gravity

Bone & muscle loss, fluid shift



Hostile closed space

Germs, sleep, noise

Astronauts must notice problems early and act on their own. Our game teaches exactly that: watch the data, decide, act.

Our solution: Collect → Assess → Act

1 Collect data

Daily health scan: heart rate, blood pressure, SpO_2, temperature, bone density, radiation dose, stress, sleep, hydration

2 Assess

Read the numbers and choose the riskiest area or the right medical decision

3 Act

Go to the right module: exercise, eat, sleep, grow plants, call family, or shelter from radiation



In-game health scan (game UI is in Turkish)

7 live health bars drop over time. If one reaches 0 → medical evacuation, mission over.

What the player experiences



A health alarm appears

Fever, heart rhythm, vision, stress or a solar storm – reach the right module in time



Make the medical call

Choose the best action and read a real space fact



Get a Health Report Card

Score, stars and every health bar at the end of the mission

Game UI shown in Turkish

Built on NASA science

Human Research Program

5 hazards (RIDGE) → our 7 health indicators and events

ISS exercise & ARED

~2 hours of exercise a day protects bone and muscle

SANS

Fluid shift → vision problems event

Immune changes

Dormant viruses can reactivate → fever event

Veggie experiment

Astronauts grow lettuce → greenhouse module

Solar particle events

Shelter + AstroRad vest tested on Artemis I

Every quiz ends with a “Real space” fact box, so players learn the real science behind each choice.

How we built it & use of AI

How it is built

- Three.js 3D graphics in one HTML file – no install
- Works offline in any browser, on PC and tablets
- Keyboard, mouse-look, first-person view, touch joystick
- Take-turns mode for up to 8 players + leaderboard

1
HTML file

0
installs

8
players

Use of AI

- AI coding assistant (Claude) helped write and test the code; the team designed the idea, content and gameplay
- Logo & team picture made with an AI image tool and marked “AI-generated image”
- No NASA logos used

Impact & what comes next

Impact

- Kids learn how astronauts stay healthy – by playing
- Teaches a real skill: read data → decide → act
- Free, offline, runs on school tablets

Next steps

- English version of the game
- Real wearable data (smartwatch heart rate)
- Difficulty levels and a Mars mission mode
- Downloadable health report

Thank you! Team AstroPulse · Azra Kurt